LANCE

An autonomous and verifiable auditor for IoT/IIoT networks

Progress report — NATO SPS G6259

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Status: work in progress.

The objective

An auditor that explores, identifies, verifies and reports.

Under which conditions
rather than which model prevails

Identical tools and budget
for both arms of the V0 control harness

Evidence, not assertion
recomputed from the log

Model training is not a current priority.

The platform in brief

A platform that runs the audit and keeps every run.

- Six phases in the LANCE pipeline, from graph analysis to report.
- Four execution modes: LANCE pipeline, rule-based audit, agent, script.
- Each run is recorded with its observations, its evidence and its cost.
- The Benchmark and Scenario Lab tabs sit beside it, on the same instance.



The platform runs the audit; the laboratory and the benchmark decide what the result means.

Captures of the running instance, 22 October 2026.

The instrument

Laboratory networks with known ground truth

Platform conducts the audit

Benchmark assesses the result

The model is one component among them.

A run constitutes a measurement only against known ground truth.

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Script, LLM agent and harness

HARNESS · fixed for both approaches: tools, budget, log, scoring

Script
A fixed procedure. The same steps run on every configuration. No model is involved.

LLM agent
A model reads each tool result and chooses the next step, within the same budget.

Tools: one shared set for both (scans, service probes, credential tests, file reads)

Log of every call, then scoring recomputed from that log

The harness is what makes the two approaches comparable: only the decision-maker changes.

Without LLM: Script and rule-based audit. With LLM: Agent simple and the LANCE pipeline.

How a run is judged against the goal

1 Goal
Find the weaknesses of a network, and only those

2 Laboratory
A scenario with planted weaknesses, listed in a ground-truth file

3 Platform
Audits the scenario: six phases, one log per run

4 Evidence
Observations and proofs saved as files for each run

5 Benchmark
Compares findings with the truth: precision, recall, F1

A result counts only against the planted truth; it never confirms the model's own conclusion.

The score is computed against the planted truth, not against the platform's own conclusion.

Precision, recall and F1 are recomputed from each run's log.

What constitutes a finding

“I have read another organisation's data.”

✕ **Not admitted**

The request and its response contain the data.

✓ **Admitted**

The same trial is conducted on a healthy and on a vulnerable network. A finding reported on the healthy one is recorded as an error.

The finding is recomputed from the log; the conclusion is never taken on trust.

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A control task before testing the model

V0 is a control task, simple enough to be solved without inference. Both approaches are required to pass.

	model ✗	model ✓
script ✓	**a defect in the instrument**	**valid — the hypotheses may be read**
script ✗	beyond the reach of this instrument	**a gain, or a coverage gap**

Only a control that the script and the model both pass permits the hypotheses to be read.

Calibration passed on the lab network (3 of 3 runs per arm). The held-out instance remains outstanding.

Current status

Pilot: both approaches succeed everywhere.

Rule-based 6 / 6



Model-based 6 / 6



The instrument does not discriminate between them.

Pilot data, 3 repetitions per family. A limitation of the instrument, not of the model.

The objective restated as hypotheses

H1 **Connected devices**

a link absent from public inputs

H2 **Heterogeneous formats**

a representation no parser covers

H3 **Novel contracts**

rights stated in unseen documentation

H4 **Vendor semantics**

a field whose meaning is only documented

H5 **Ambiguous refusals**

resuming the audit after a refusal

H6 **Hostile content**

an instruction obeyed from observed content

H1–H5 quantify a possible gain. **H6 quantifies a risk.**

The register is this list of hypotheses and the test of each.

What we cannot claim

✕	**No general advantage**	No hypothesis has yet been tested over repeated runs.
✕	**No confidence interval**	Three repetitions yield a dispersion, not a sample.
✕	**No admitted evidence**	The metric definition is not frozen.

A successful run is not a measurement: one early run reported success while every model arm had failed.

Next steps

1. Execute V0 on a held-out instance
2. Develop a parameterisable test harness
3. Build the scenario families for H2, H3 and H4
4. Deploy a reproducible platform

Support requested

- access to real vendor artifacts
- an adversarial review of the register

Order follows conditions, not dates.

AsiaCCS 2027 — submission deadline 11 December 2026 (AoE).

Reserve — declared coverage gaps

- Physical impact and functional safety
- Device fragility under probing
- Real fleet scale
- Proprietary industrial protocols that are not simulated
- Complete multi-step exploitation chains
- Abstention and calibration
- Generic IT scenarios, deliberately excluded

A benchmark that states its gaps is stronger than one that claims full coverage.